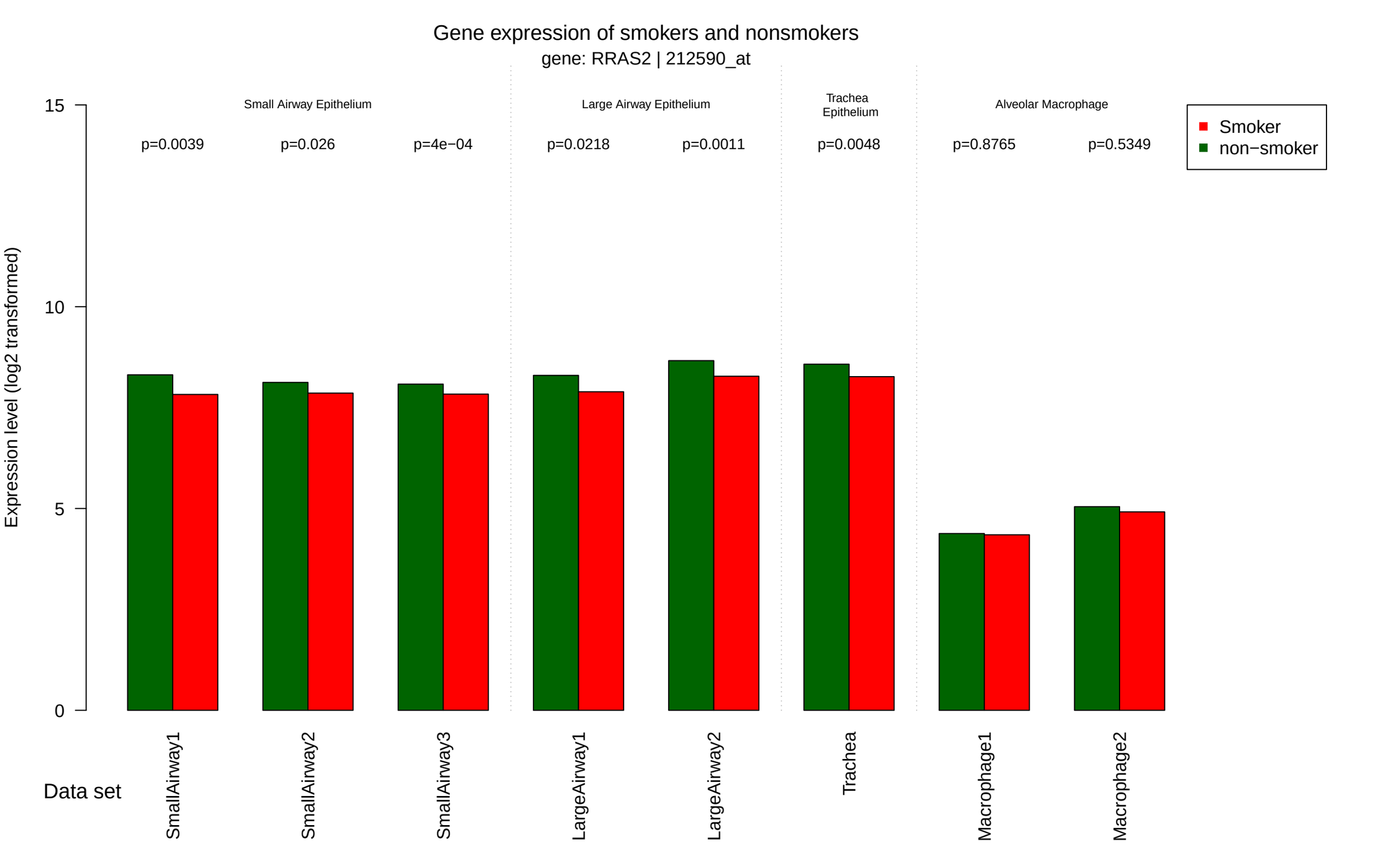
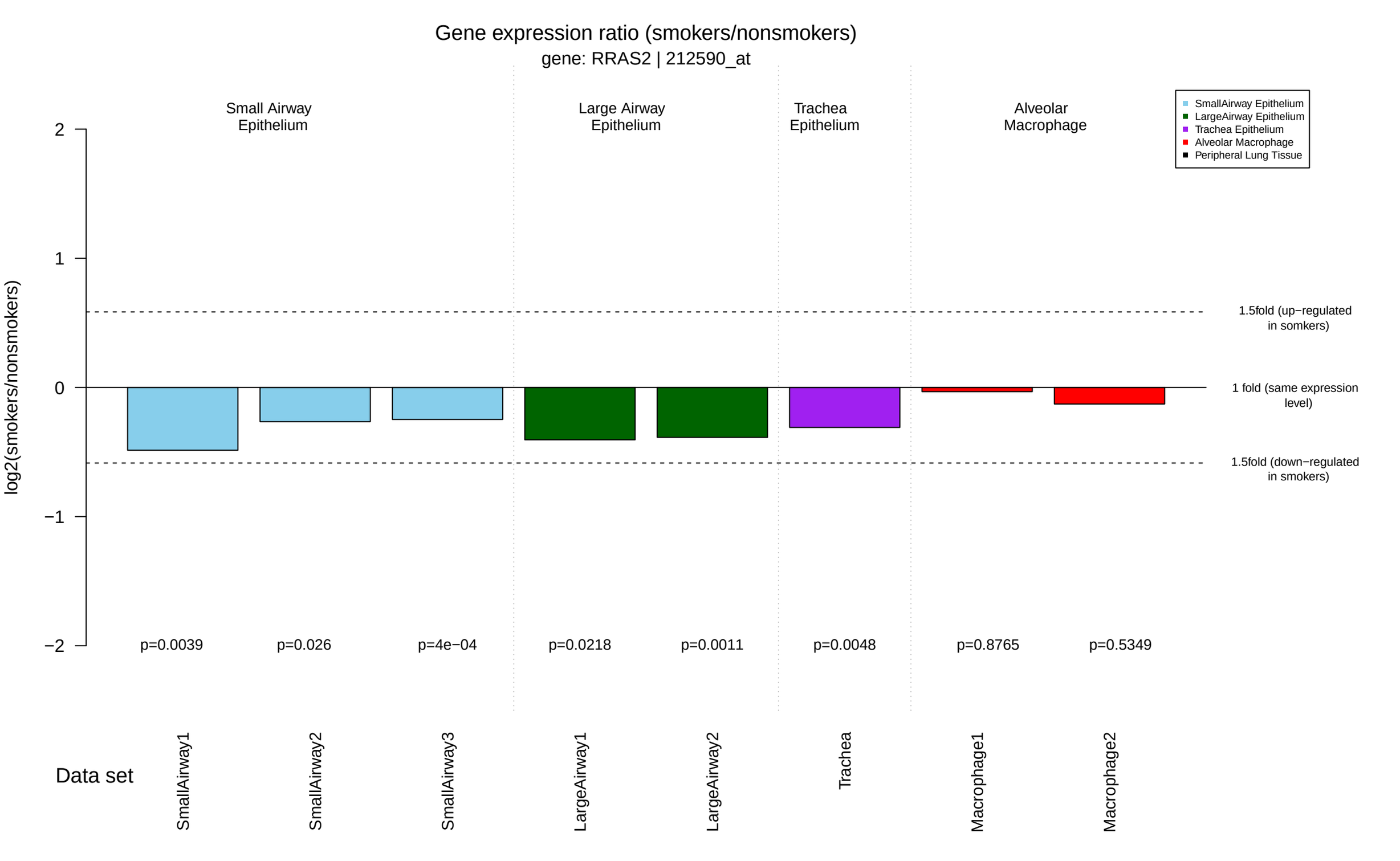
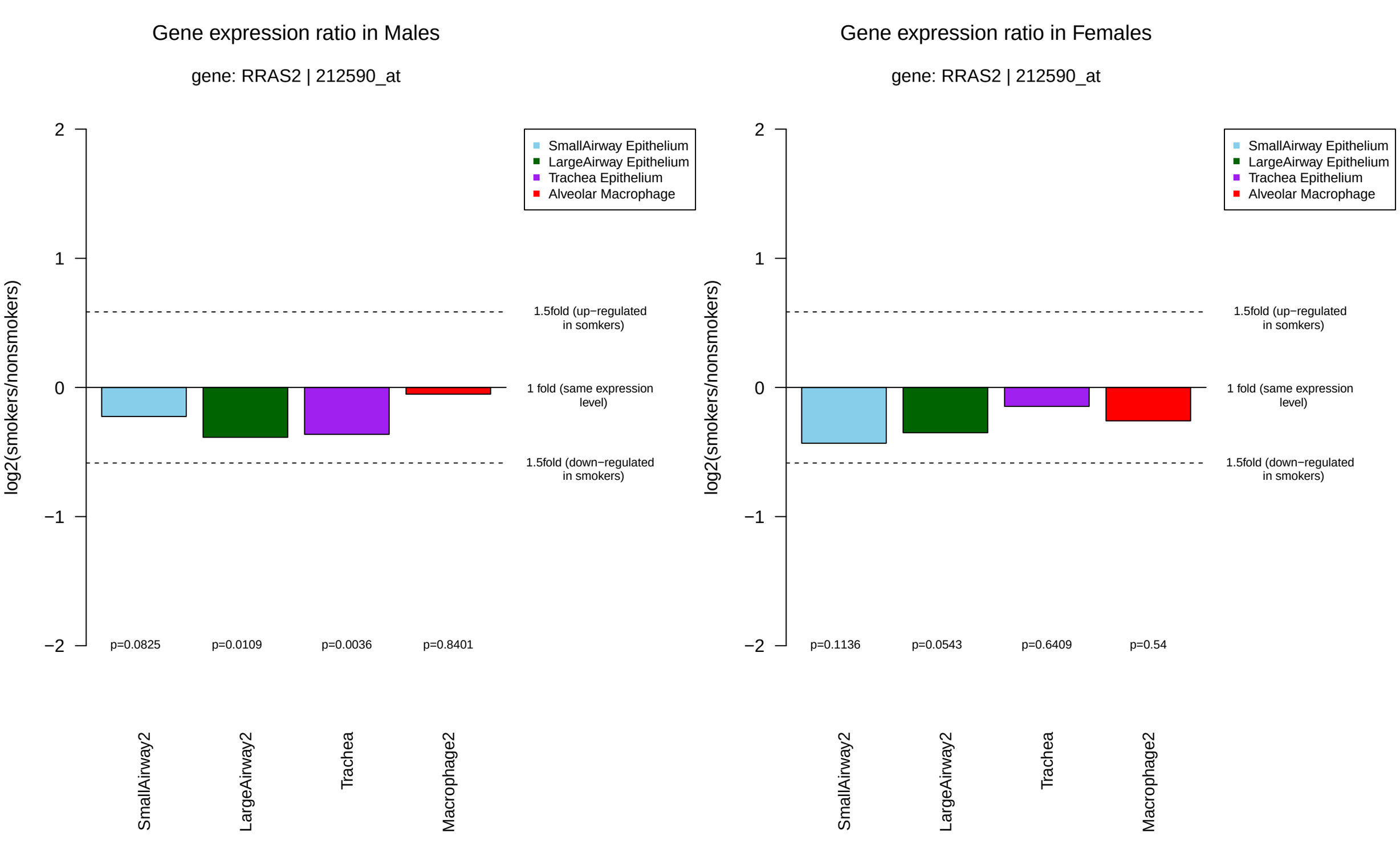
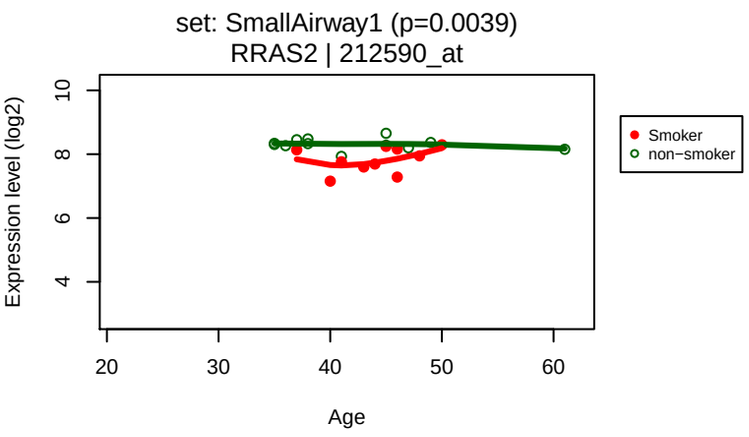
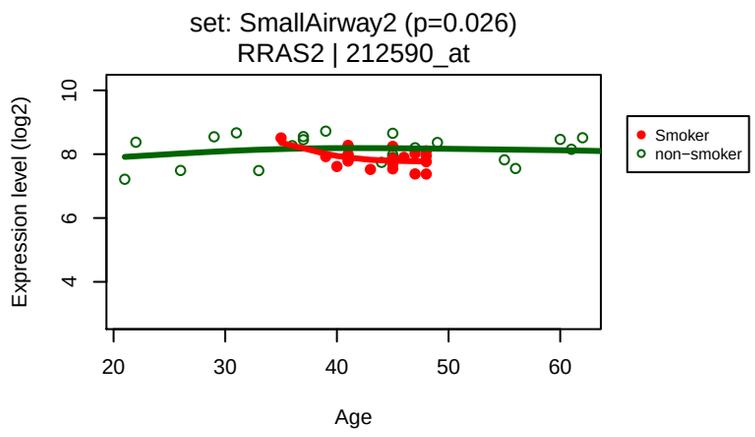
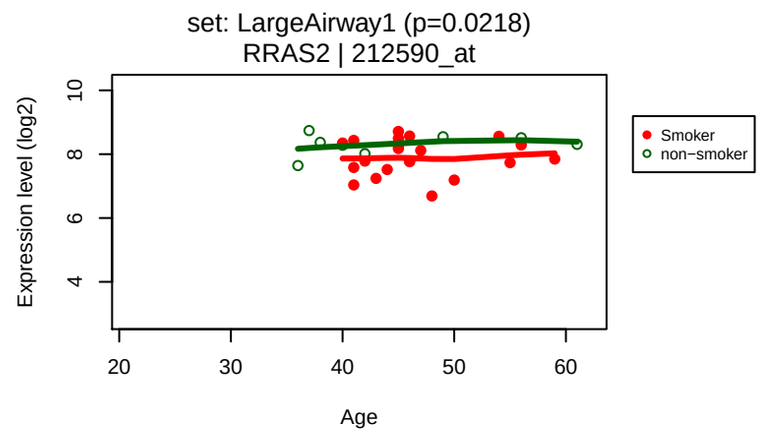

Gene expression of smokers and nonsmokers
gene: RRAS2 | 212590_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0039
p=0.026
p=4e−04
p=0.0218
p=0.0011
p=0.0048
p=0.8765
p=0.5349
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

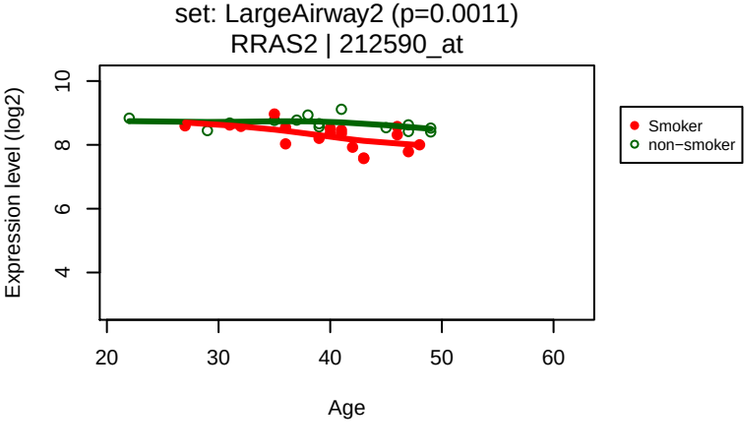

Gene expression ratio (smokers/nonsmokers)
gene: RRAS2 | 212590_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0039
p=0.026
p=4e−04
p=0.0218
p=0.0011
p=0.0048
p=0.8765
p=0.5349
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

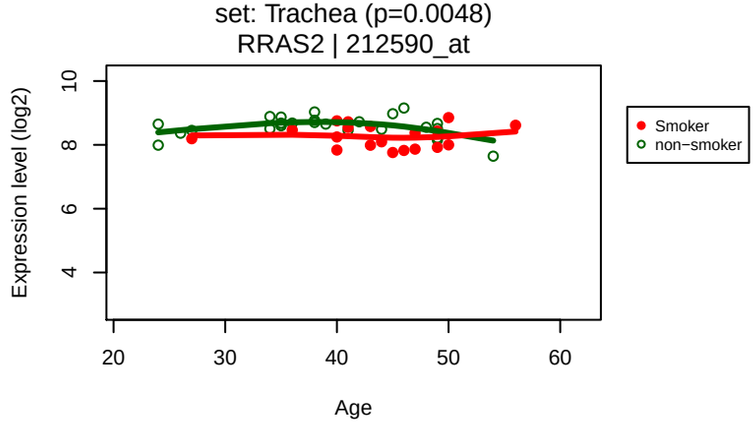

Gene expression ratio in Males
gene: RRAS2 | 212590_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0825
p=0.0109
p=0.0036
p=0.8401
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RRAS2 | 212590_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1136
p=0.0543
p=0.6409
p=0.54
SmallAirway2
LargeAirway2
Trachea
Macrophage2

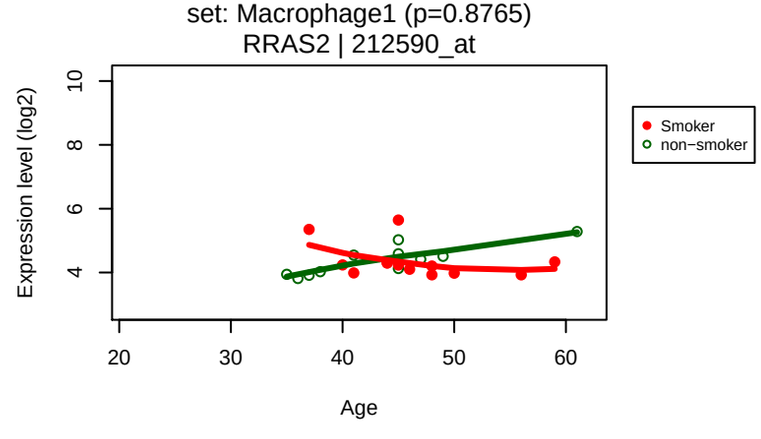

set: SmallAirway1 (p=0.0039)
RRAS2 | 212590_at
Expression level (log2)
Age
Smoker
non−smoker

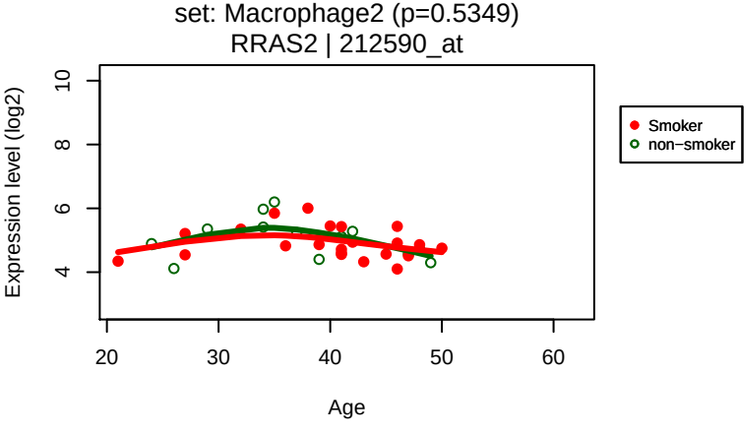

set: SmallAirway2 (p=0.026)
RRAS2 | 212590_at
Expression level (log2)
Age
Smoker
non−smoker

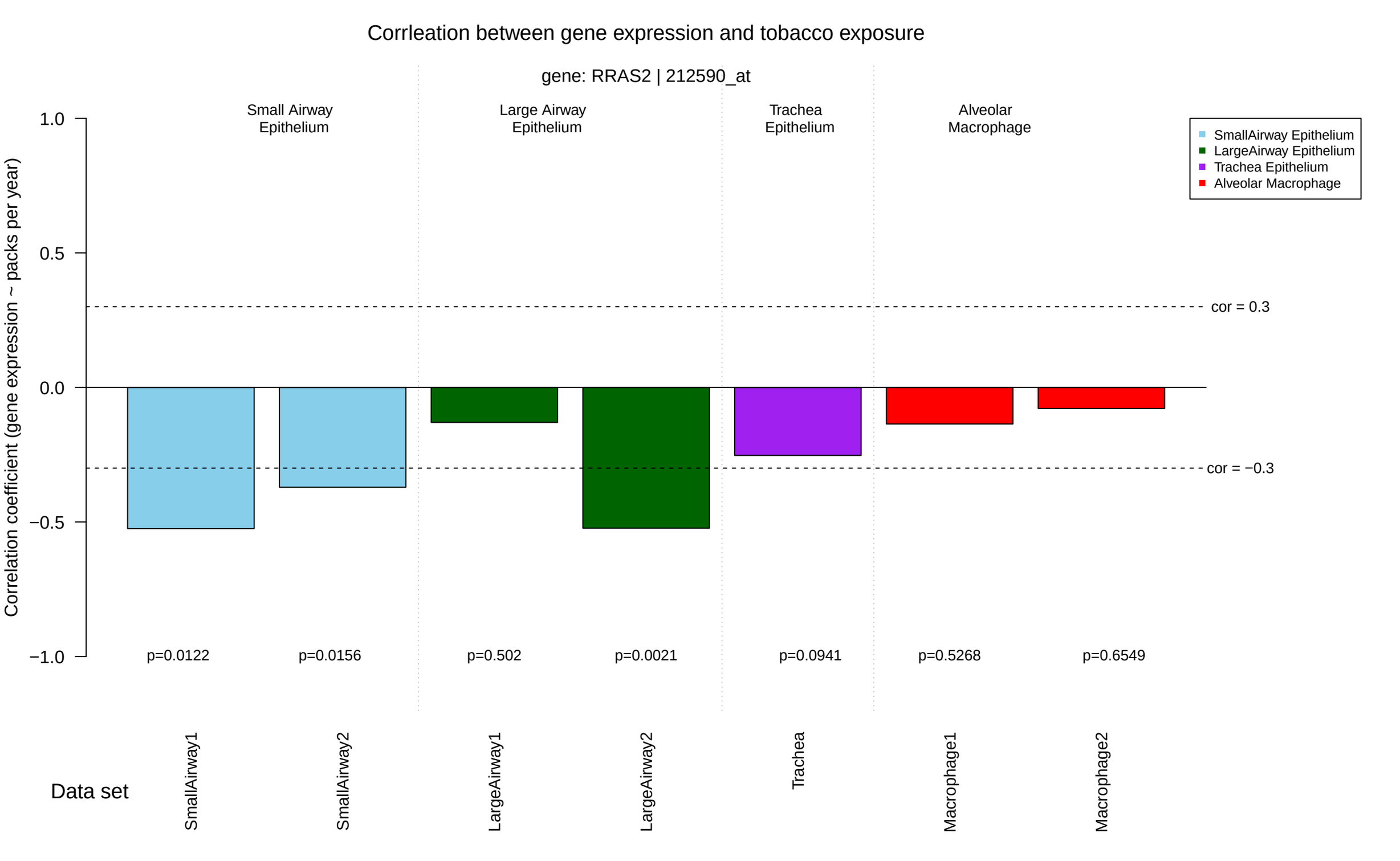

set: LargeAirway1 (p=0.0218)
RRAS2 | 212590_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0011)
RRAS2 | 212590_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0048)
RRAS2 | 212590_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8765)
RRAS2 | 212590_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5349)
RRAS2 | 212590_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RRAS2 | 212590_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0122
p=0.0156
p=0.502
p=0.0021
p=0.0941
p=0.5268
p=0.6549
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2